SATQUERY AI — REMOTE SENSING MISSION REPORT

Indian Space Research Organisation (ISRO) | SIH 26167 Intelligence Dispatch

Natural Language Query:	Quantify urban sprawl, new construction, and vegetation loss between 2022 and 2024.
Identified Workflow Intent:	CHANGE_DETECTION
Dispatched Specialist:	ChangeDetectionEngine
Composite Confidence:	**94.8% (HIGH)**

1. Executive Analysis & Findings

Assessment: Bi-temporal comparative analysis reveals significant anthropogenic change: Built-up impervious surface expanded by +18.7% (approx. 6.97 sq km), primarily concentrated in eastern logistics and residential corridors. Vegetation canopy exhibited a concurrent decrease of -12.4% due to land clearing.

- Built-up area expansion: +18.7% over the observation interval
- Vegetation cover reduction: -12.4% (Agricultural/Woodland to Urban)
- Spatial co-registration validated with IoU >= 85%

2. 4-Signal Multi-Criteria Confidence Matrix

Metric Signal	Component	Score (%)	Status
C_model	Specialist Model Inference Certainty	94.0%	Optimal
C_sensor	Sensor Radiometric SNR & Quality	90.0%	Optimal
C_alignment	Spatial Co-Registration (CRS & IoU)	100.0%	Verified
C_resolution	Ground Sample Distance Suitability	95.0%	Optimal

3. Auditable Agent Execution Pipeline Trace

Step	Action / Milestone	Tool Executed	Latency	Status
1	Raster Ingestion & Metadata Parsing	GeoTIFFProcessor	1.6 ms	COMPLETED
2	Spatial Co-Registration & Alignment Vali	SpatialRegistrationValidator	1.2 ms	COMPLETED
3	Agentic Query Classification & Tool Rout	AgentClassifier	0.0 ms	COMPLETED
4	Specialist Model Execution (ChangeDetect	ChangeDetectionEngine	0.0 ms	COMPLETED
5	Evidence Fusion & 4-Signal Confidence Ra	CompositeConfidenceEngine	0.0 ms	COMPLETED